English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Year 3, Stream 2 Written examination, 2 July 2023

Problem 1. A speed-measuring device's percentage error has distribution $N(0, 5^2)$.

1. (0.25 points) On average, how many cars pass before the first error of at least 10%? Among 1000 cars, how many errors exceed 10% on average? After 100 cars, what is the probability that the mean error exceeds 1%?
2. (0.25 points) Model passing speeds in km/h as independent $N(40, 5^2)$ variables. What is the probability that the device records a speed above the 50 km/h limit? What must its standard deviation be for the mean error after 100 cars to exceed 1 km/h with probability 0.1%?
3. (0.5 points) Answer part 1 using absolute error, the modulus of the error in that part.

Problem 2. A travel company advertises seaside hotels by repeatedly calling $N \sim \text{Poi}(\lambda)$ contacts. Person i accepts after K_i calls, where the K_i are iid, independent of N, with $\mathbb{P}(K_i = n) = f(n)$. The value ∞ is allowed, so $f(\infty) \geq 0$. Let $X_n = \sum_{i=1}^{N} \mathbb{1}_{\{K_i = n\}}$ count holidays sold in round n.

1. (0.5 points) Prove that the X_n are independent and $X_n \sim \text{Poi}(\lambda f(n))$.
2. (0.5 points) The company stops after $T = \inf\{n : X_n = 0\}$, the first round with no new customer. Let $S = X_1 + \cdots + X_T$. Prove $\mathbb{E}[S] = \lambda \mathbb{E}[F(T)]$, where $F(k) = f(1) + \cdots + f(k)$.

Problem 3. Let $U \sim U(0, 1)$ and $V \sim U(-\pi/2, \pi/2)$.

1. (0.5 points) One method of generating a pseudorandom number between 0 and 1 from U chooses a usually large positive integer M and takes $X = \{MU\}$, the fractional part. Find $\mathbb{E}X$, DX and $\text{Cor}(U, X)$. What do you think of this method?
2. (0.5 points) Let $U_1, \ldots, U_{100}$ be iid copies of U. Find their individual expectations and those of the maximum and minimum.
3. (0.5 points) Find the density, expectation and variance of $Y = \tan V$.
4. (0.5 points) For $Z \sim \text{Cauchy}(1)$ with density $1/[\pi(1 + x^2)]$, prove that $1/Z$, $2Z/(1 - Z^2)$ and $(3Z - Z^3)/(1 - 3Z^2)$ have the same distribution.

Bonus

1. (0.25 points) Define $D(Y \mid X) = \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 \mid X]$. Recall $\mathbb{E}X = \mathbb{E}[\mathbb{E}[X \mid Y]]$. Which of

$$\mathbb{E}[D(Y \mid X)], \quad D(D(Y \mid X)), \quad \mathbb{E}[D(Y \mid X)] + D(D(Y \mid X)),$$

$$\mathbb{E}[D(Y \mid X)] + D(\mathbb{E}[Y \mid X])$$

expresses DY in terms of conditional variance?

Construct Cantor's function $c : [0, 1] \to [0, 1]$ recursively: $f(x) = 1/2$ on $[1/3, 2/3]$, $f(x) = f(3x)/2$ on $[0, 1/3)$, and $f(x) = (1 + f(3x - 2))/2$ on $(2/3, 1]$.

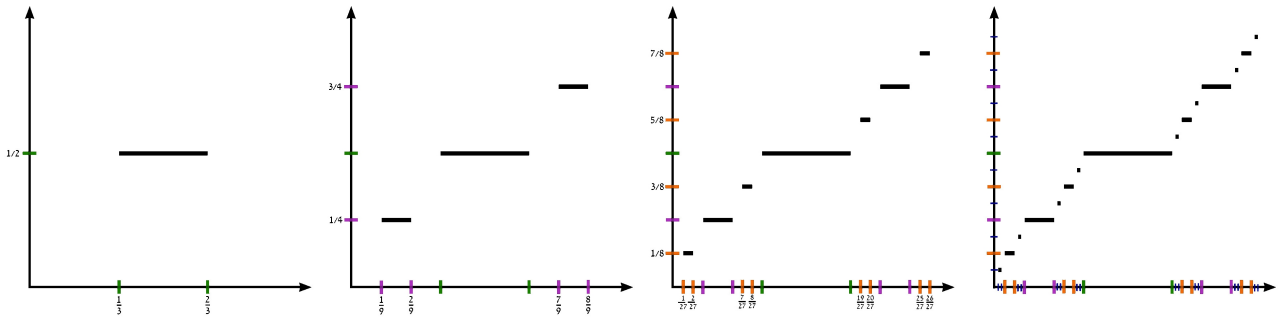


Figure 1: Recursive construction of Cantor's function. Source: Wikipedia.

2. (0.25 points) Let $F_X(x) = c(x)$ for $x \in (0, 1)$. Find $\mathbb{E}X$ and DX.
3. (0.25 points) Prove that F_X is differentiable almost everywhere. What is the density f_X?